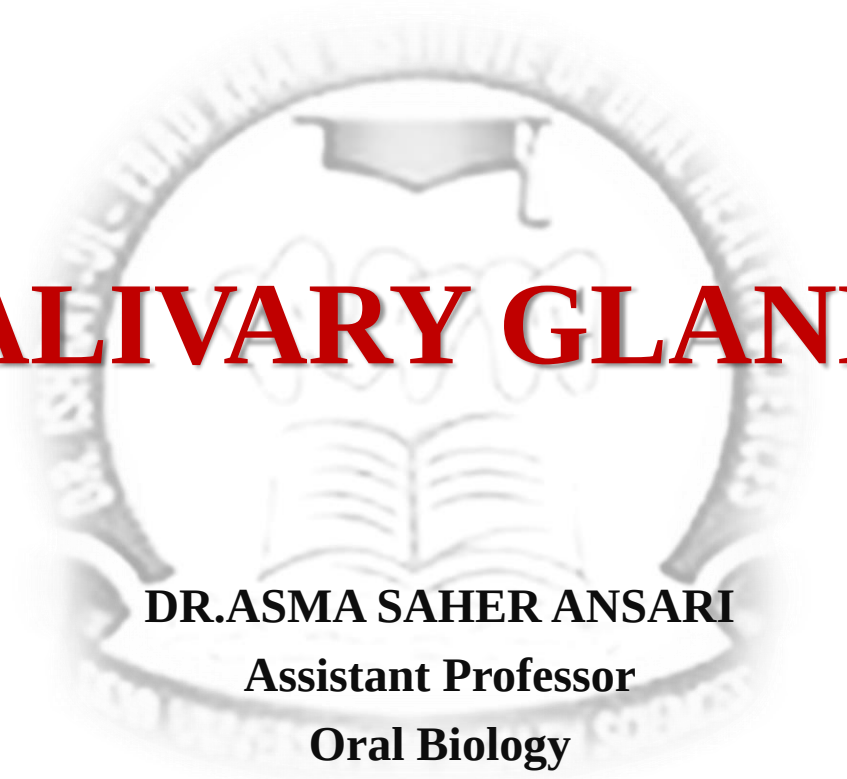




SALIVARY GLANDS

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DIKIOHS,DUHS

OBJECTIVES

By the end of the lecture you will be able to understand:

1. What are salivary glands and what is saliva?
2. Function and composition of saliva
3. Mechanism of salivary secretion.
4. Clinical correlations.

Introduction

SALIVA:

- **Complex fluid produced by salivary glands.**
- **Keep oral cavity moist**
- **Maintains well being of the oral cavity.**

COMPOSITION OF SALIVA

<i>PARAMETER</i>	CHARACTERISTICS
<i>volume</i>	600-1000ML/DAY
<i>electrolytes</i>	Na ,K, Cl, HCO, Mg, HPO, SCN & F
<i>secretory proteins</i>	Amylase, proline-rich proteins, mucins, histatin, cystatin, peroxidase, lysozyme &lecoferrin
<i>immunoglobulins</i>	IgA, IgG, IgM

COMPOSITION OF SALIVA cont....

Small organic molecules

Glucose, amino acid, urea, uric acid & lipid

Other components

EGF, insulin, cyclic adenosine monophosphate-binding proteins & serum albumin

FLOW RATE OF SALIVA

*Flow rate
(ml/min)*

Whole

Resting

0.2-0.4

Stimulated

2.0-5.0

pH

6.7

SALIVARY GLANDS

Salivary glands may be classified according to size (major and minor) and/or the types of secretion (mucous, serous or mixed).

MAJOR SALIVARY GLANDS:

3 pairs, located outside the mouth.

- Parotid
- Submandibular
- Sublingual

These glands have an extended duct system through which saliva reaches the mouth.

Parotid salivary gland

Parotid duct

Masseter muscle

Mucosa (cut)

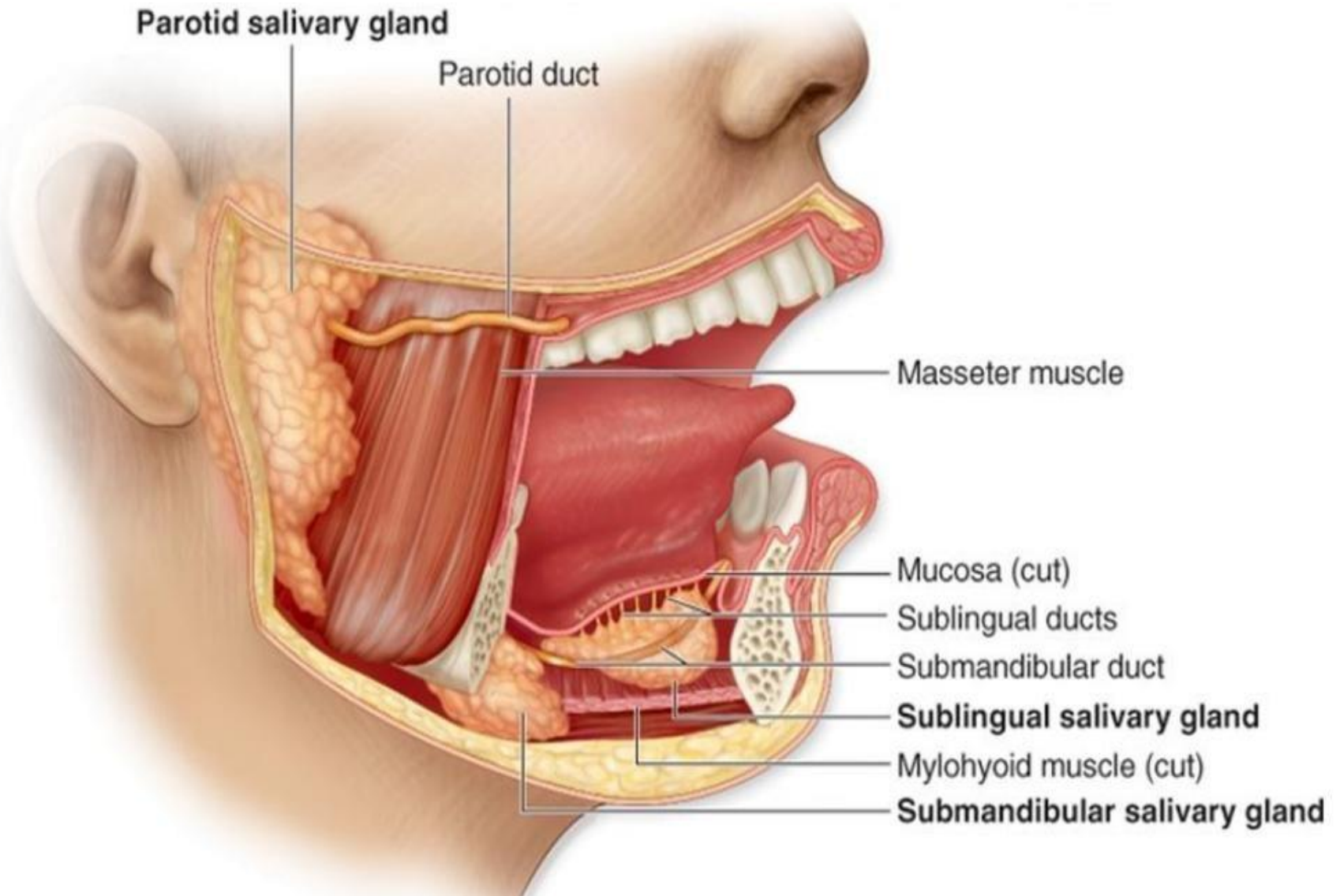
Sublingual ducts

Submandibular duct

Sublingual salivary gland

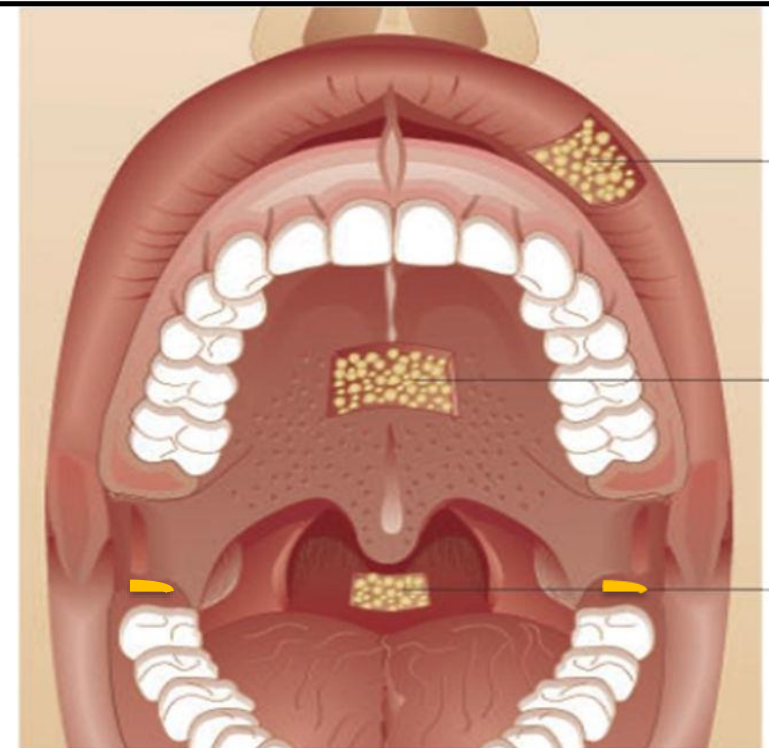
Mylohyoid muscle (cut)

Submandibular salivary gland



SALIVARY GLANDS

- **Minor salivary glands:**
 - Numerous
 - Located in various parts of oral cavity within submucosa.
 - lingual, labial buccal, palate, pharyngeal & retromolar region.
 - Have short ducts open directly on the mucosal surface.



Salivary glands are not present in the gingiva or the dorsum of the anterior two-thirds of the tongue.

Functions of saliva

FUNCTION	EFFECT	ACTIVE CONSTITUENTS
1. Protection	• Clearance • Lubrication • Thermal/chemical insulation • Pellicle formation	Water Mucin&glycoproteins mucins Proteins, glycoproteins & mucins

FUNCTIONS OF SALIVA

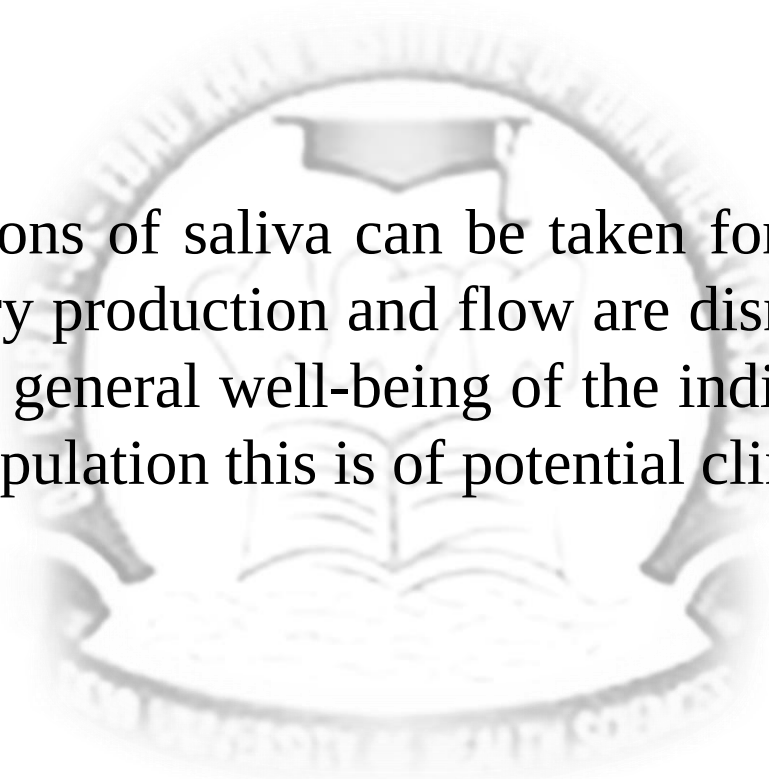
FUNCTION	EFFECT	ACTIVE CONSTITUENTS
2. Buffering	- pH maintenance - Neutralization of acid	Bicarbonates & phosphate Basic protein,
3. Tooth integrity	Enamel maturation & repair	Ca, phosphate, fluoride, statherin & proline-rich proteins

FUNCTIONS OF SALIVA

FUNCTION	EFFECT	ACTIVE CONSTITUENT
4.Antimicrobial activity	• Physical barrier • Immune defence • Nonimmune defence	Mucins IgA Peroxidase, lysozyme, lactoferrin,histatin, mucins & agglutinins
5.Tissue repair	• Wound healing & epi. regeneration	Growth factors & terofil proteins

FUNCTIONS OF SALIVA

FUNCTION	EFFECT	ACTIVE CONSTITUENT
6.Digestion	• Bolus formation • Starch & triglyceride digestion	Water & mucins Amylase & lipase
7.Taste	• Solution of molecules • Maintenance of taste buds	Epidermal growth factor & carbonic hydrase VI



The varied functions of saliva can be taken for granted and it is only when salivary production and flow are disrupted that its true importance to the general well-being of the individual is realized: with an ageing population this is of potential clinical importance .

NATURE OF SALIVA

- **Parotid gland** secrete a watery saliva rich in enzymes such as amylase, proteins(e.g.,proline-rich protein) other glycoproteins.
- **Submandibular** saliva contains highly glycosylated substance called mucins.
- **Sublingual gland** produces a viscous saliva rich in mucins.

STRUCTURE OF SALIVARY GLAND

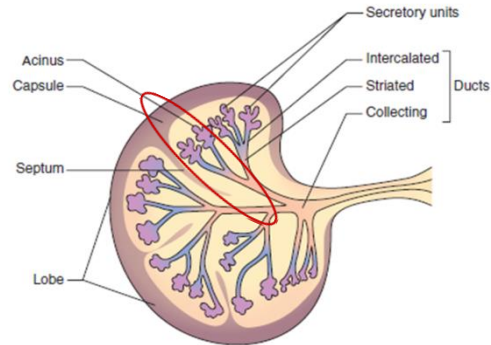
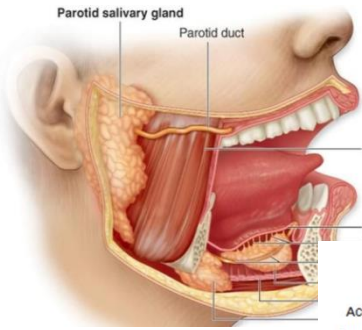
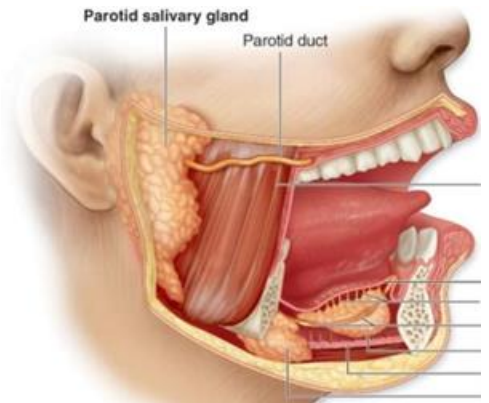
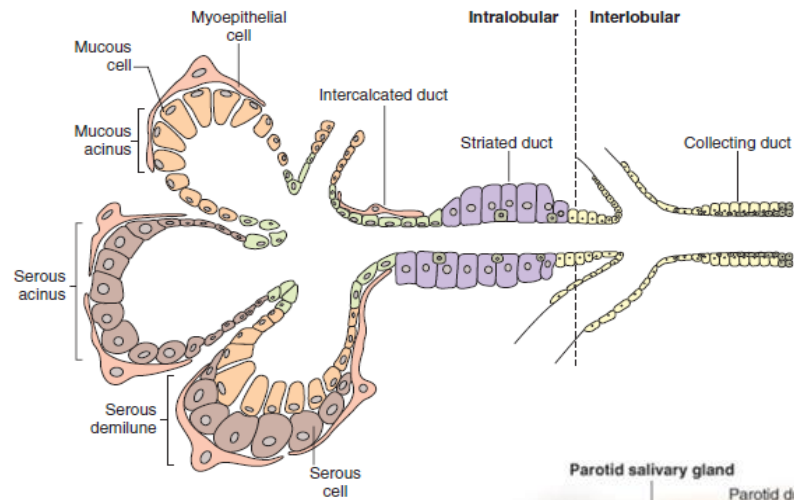
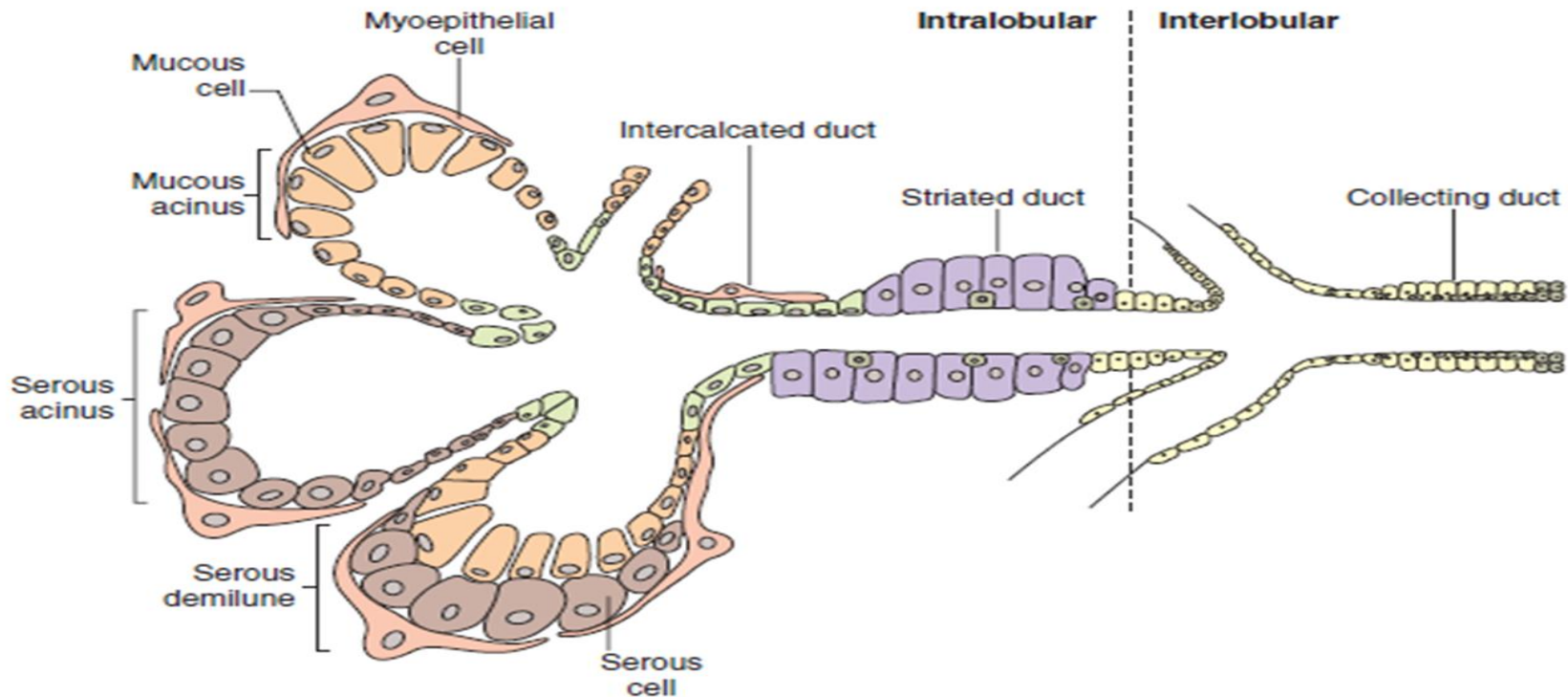


Fig. 16.1 The general organization of a salivary gland.



The acinus, via its lumen, empties into an intercalated duct lined with cuboidal epithelium, which in turn joins a larger striated duct formed of columnar cells.



Both *the intercalated and striated ducts* are *intralobular* and affect the composition of the secretion passing through them.

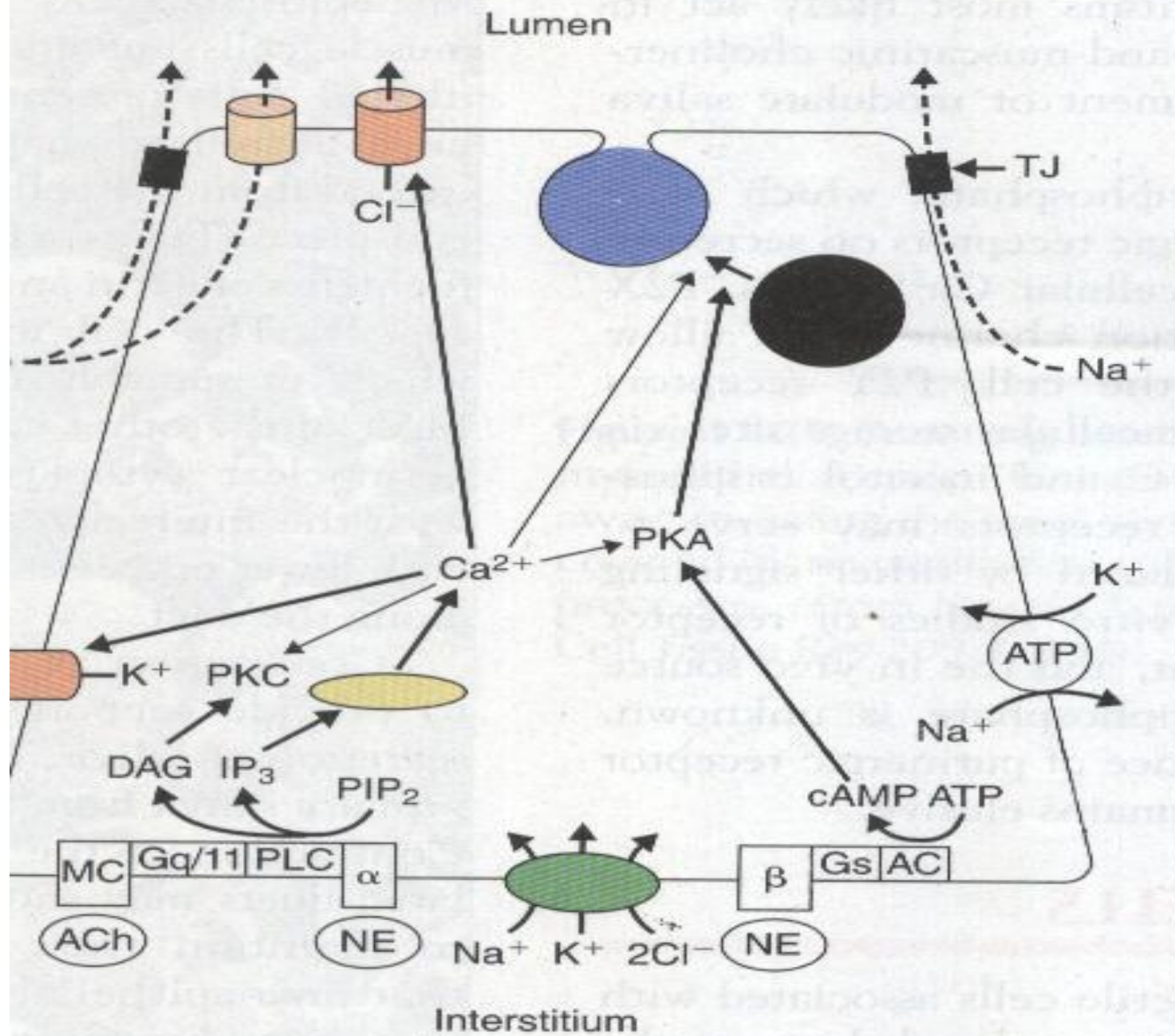
The *striated ducts* empty into the collecting ducts, which are mainly *interlobular*.

FORMATION & SECRETION OF SALIVA

- Formation of saliva occurs in two stages:
- First stage,
cells of secretory end pieces & intercalated ducts produce primary saliva (isotonic) containing most of the organic components & all of the water that is secreted by their salivary glands.

FORMATION & SECRETION OF SALIVA

- Second stage, primary saliva is modified as it passes through the striated & excretory ducts, mainly by reabsorption & secretion of electrolytes.
- The saliva that reaches the oral cavity is hypotonic.



Other mechanisms modulating saliva secretions

- Substance P, vasoactive intestinal polypeptide, neuropeptide Y & calcitonin gene related peptide effects the glandular vasculature to regulate blood flow.

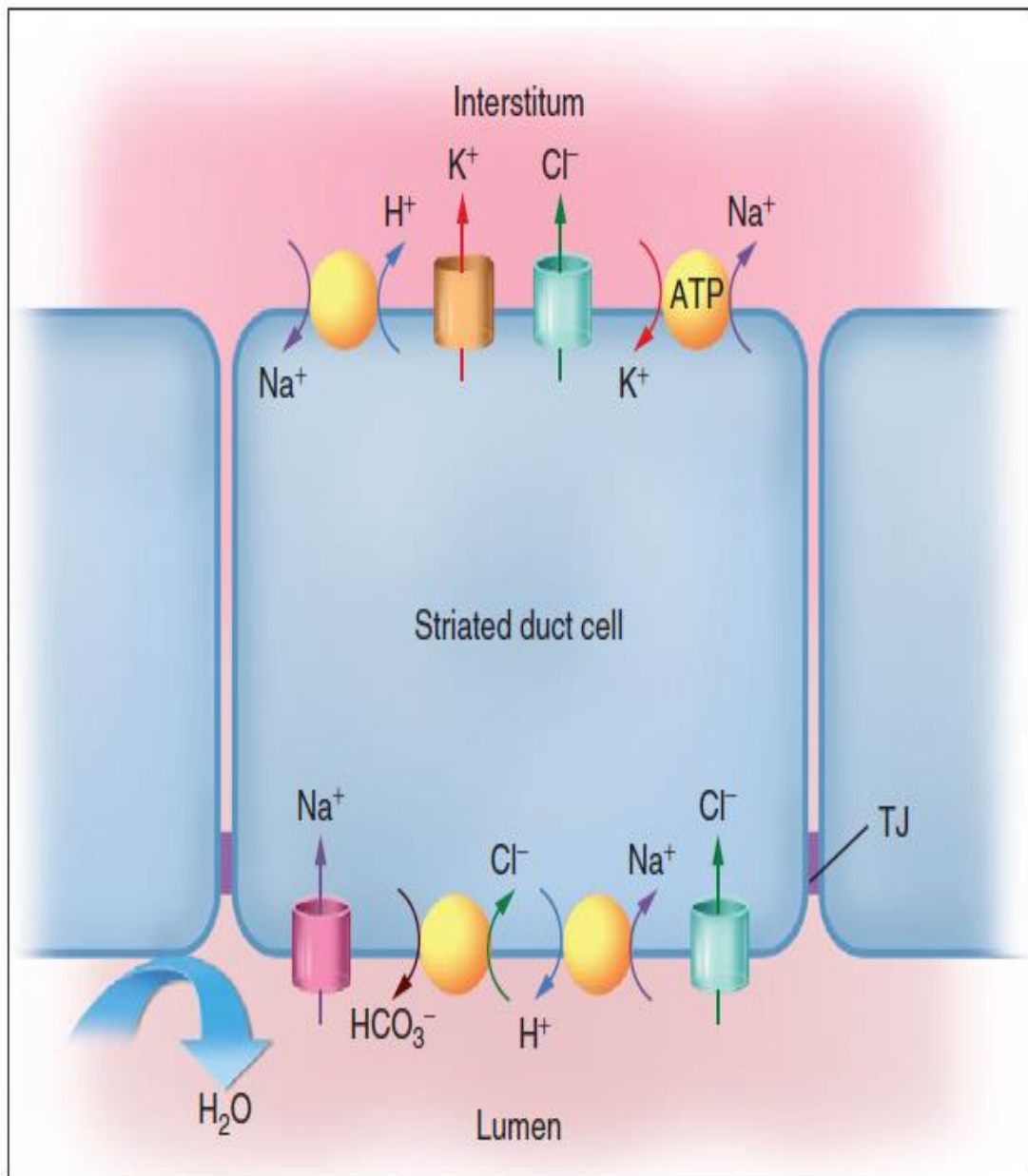
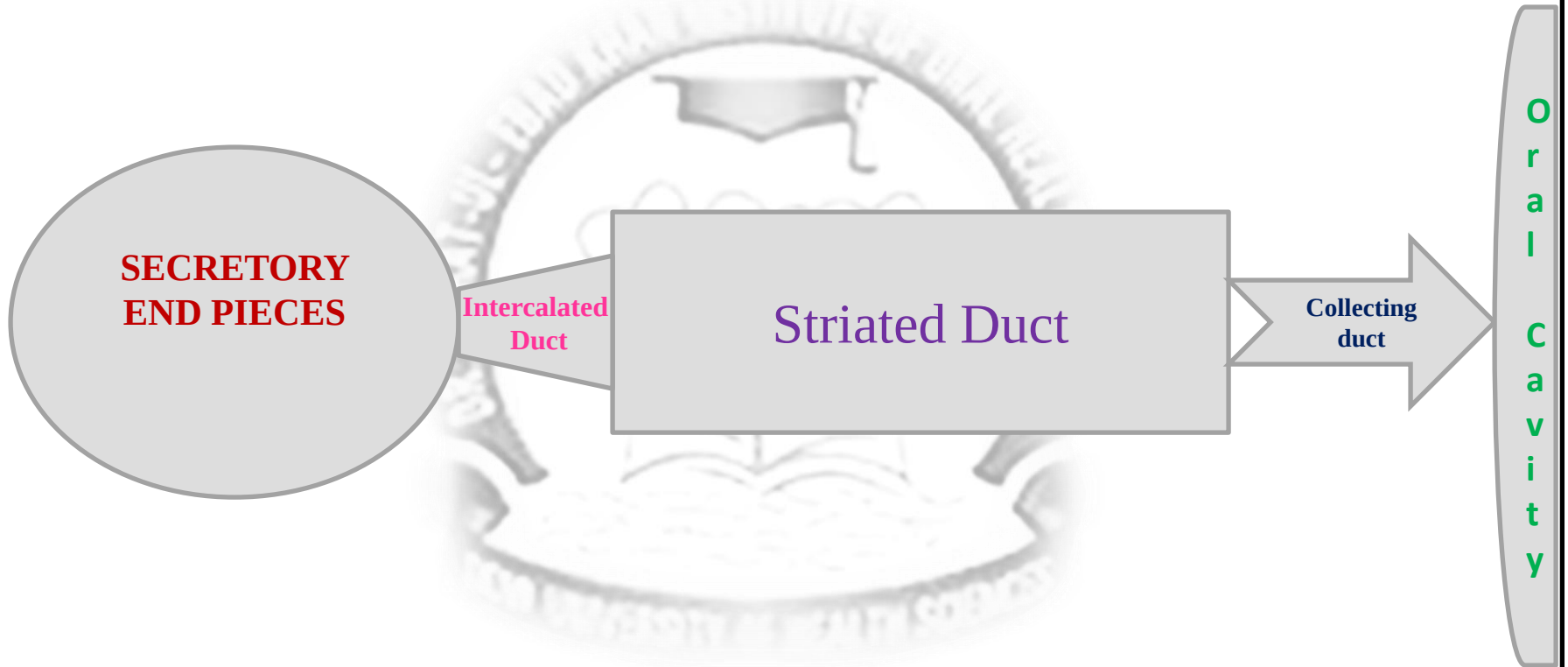


FIGURE 11-31 Mechanisms of ductal modification of saliva. Striated duct cells reabsorb Na⁺ and Cl⁻ mainly via channels in the luminal membrane; Na⁺/H⁺ and Cl⁻/HCO₃⁻ exchangers provide additional mechanisms for uptake of these ions. Na⁺ exits at the basolateral surface via the Na⁺/K⁺-adenosinetriphosphatase, and Cl⁻ exits via a channel. K⁺ channels at the basolateral surface maintain electroneutrality, and the Na⁺/H⁺ exchanger compensates for intracellular acidification. The tight junctions (TJ) are relatively “tight,” and the duct cells are impermeable to water. (Modified from Melvin JE: *Crit Rev Oral Biol Med* 10:199, 1999.)

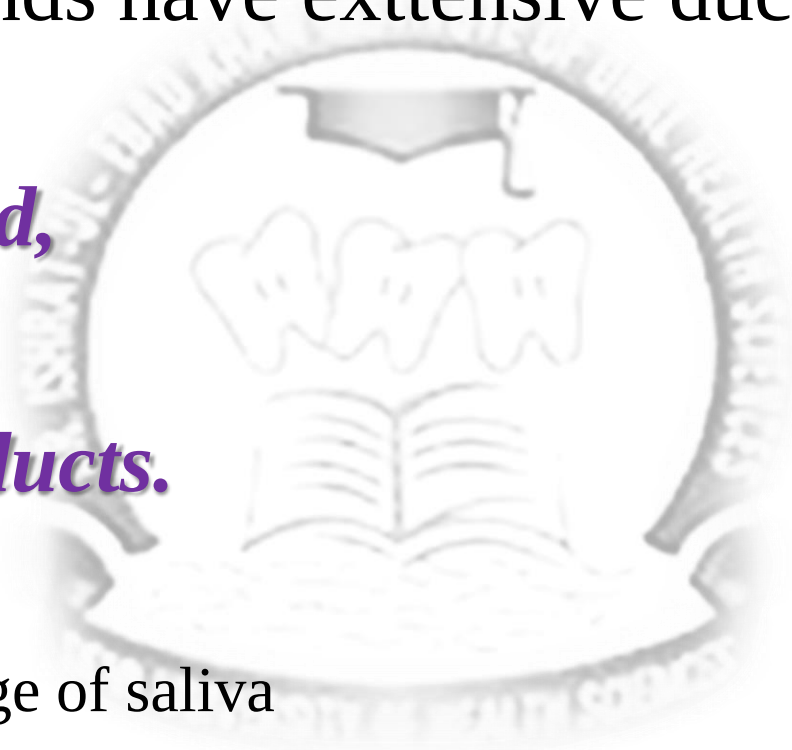
STRUCTURE OF SALIVARY GLAND



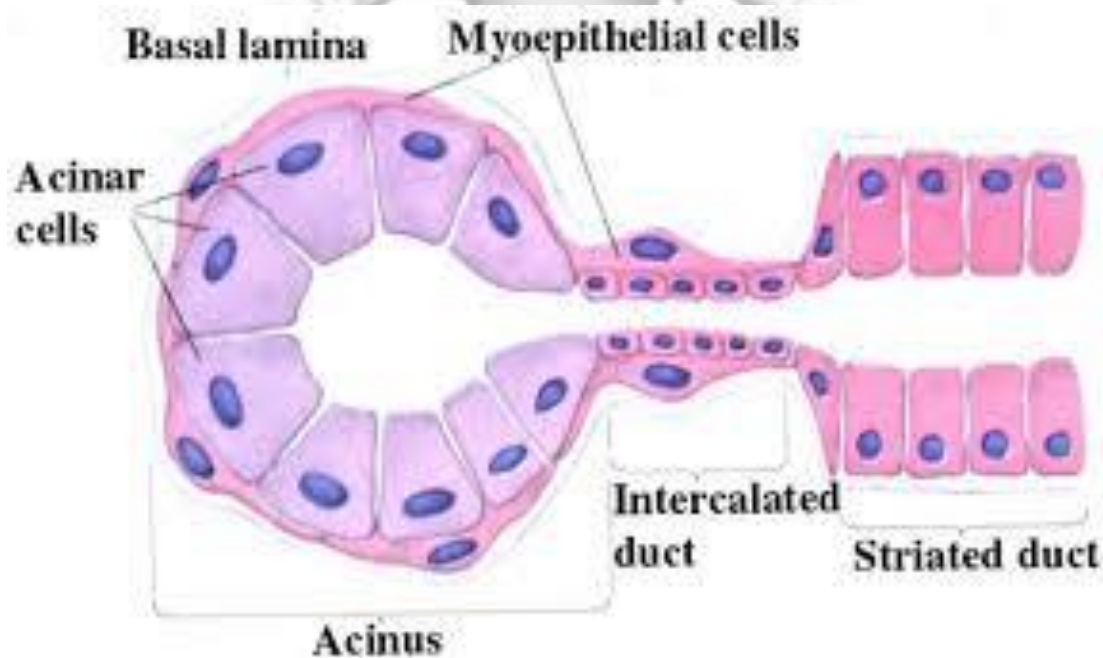
DUCTS

Salivary glands have extensive duct system, it consist of:

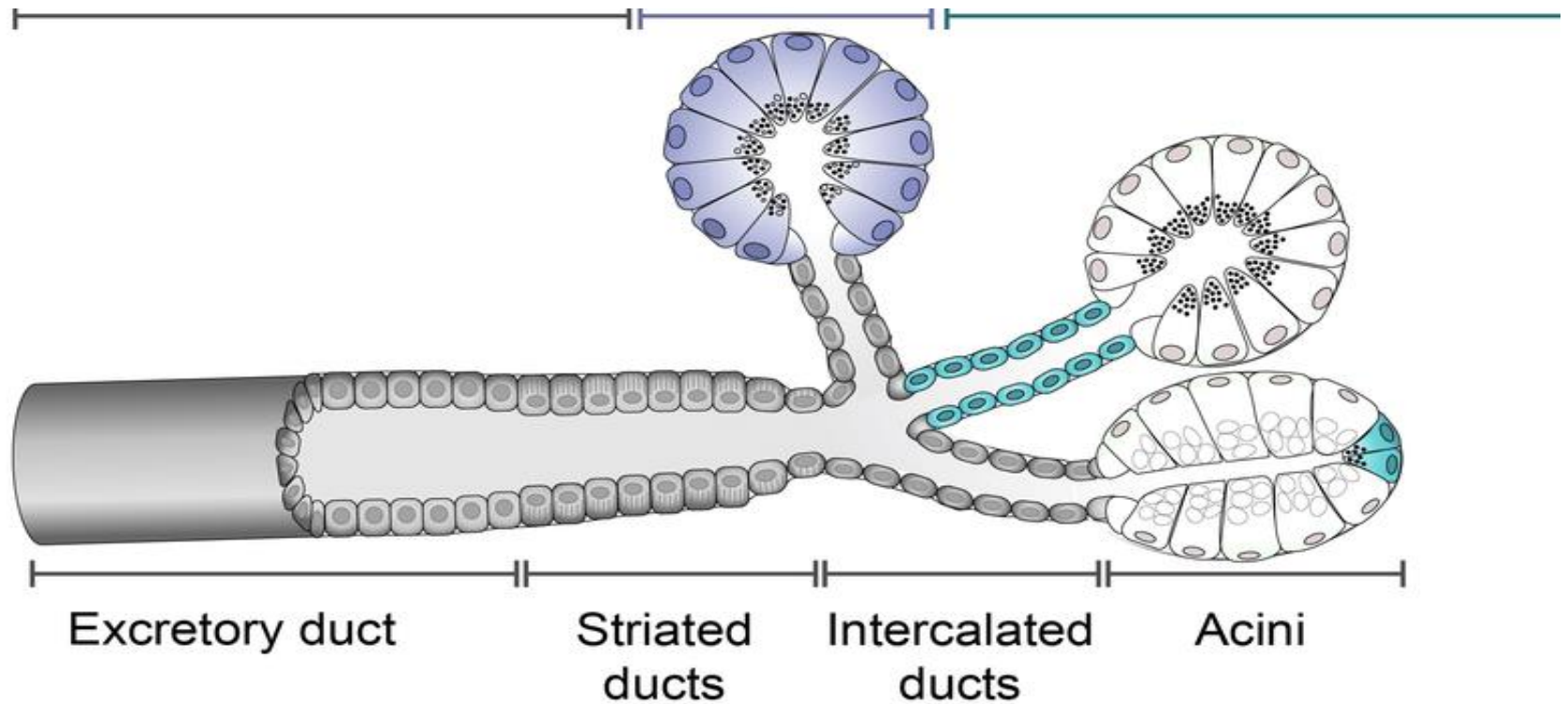
- *intercalated,*
 - *striated &*
 - *excretory ducts.*
-
- Act as a passage of saliva
 - Also actively participates in the production & modification of saliva.



- PRIMARY Saliva produced by the secretory end pieces passes first thru these ducts.
- The lumen of end piece is continuous with the lumen of intercalated duct.



Lined by simple cuboidal epithelium.
myoepithelial cells & their processes are typically located along basal surface of the duct



Overall dia. Of duct is greater than that of secretory end pieces, the lumen is larger than those of secretory end pieces & intercalate ducts.

Modification of primary saliva by reabsorption & secretion of electrolytes.

Modification of saliva occurs thru reabsorption & secretion of electrolytes.

Luminal & basolateral membranes have abundant transporters .

The transporters produce a net absorption of Na & Cl, resulting in the formation of hypotonic final saliva.

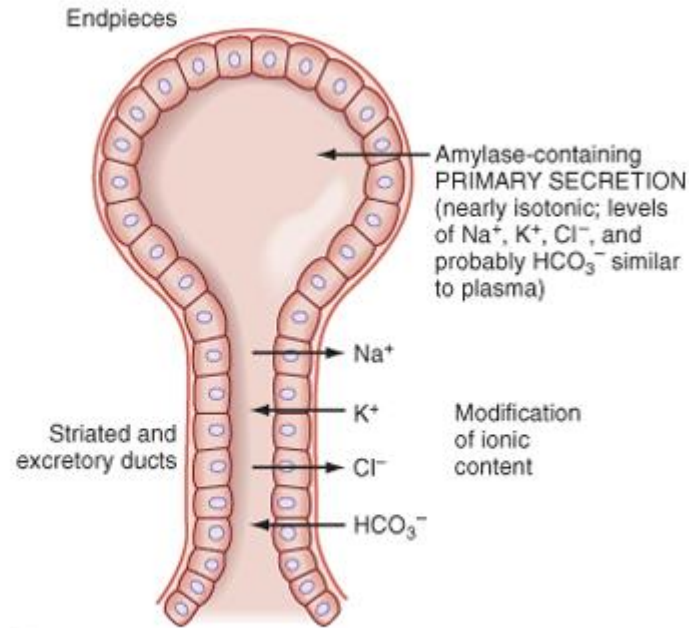
Ducts also secrete K & HCO₃.

The final electrolyte composition of saliva varies , depending on the salivary flow rate.

At low flow rates, HCO₃ conc. Increases.

Electrolyte reabsorption & secretion by the striated & excretory ducts is regulated by ANS & by mineralocorticoids produced by the adrenal cortex.

A



B

Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.
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- At high flow rates saliva is in contact with ductal epithelium for a shorter time & K conc. decreases.
- At low flow rates, HCO_3 conc. Increases.
- Electrolyte reabsorption & secretion by the striated & excretory ducts is regulated by ANS & by mineralocorticoids produced by the adrenal cortex.

- Striated duct cells reabsorb Na^+ and Cl^- mainly via channels in the luminal membrane;
- Na^+/H^+ and $\text{Cl}^-/\text{HCO}_3^-$ exchangers provide additional mechanisms for uptake of these ions.
- Na^+ exits at the basolateral surface via the Na^+/K^+ adenosinetriphosphatase, and Cl^- exits via a channel.
- K^+ channels at the basolateral surface maintain electroneutrality, and the Na^+/H^+ exchanger compensates for intracellular acidification.

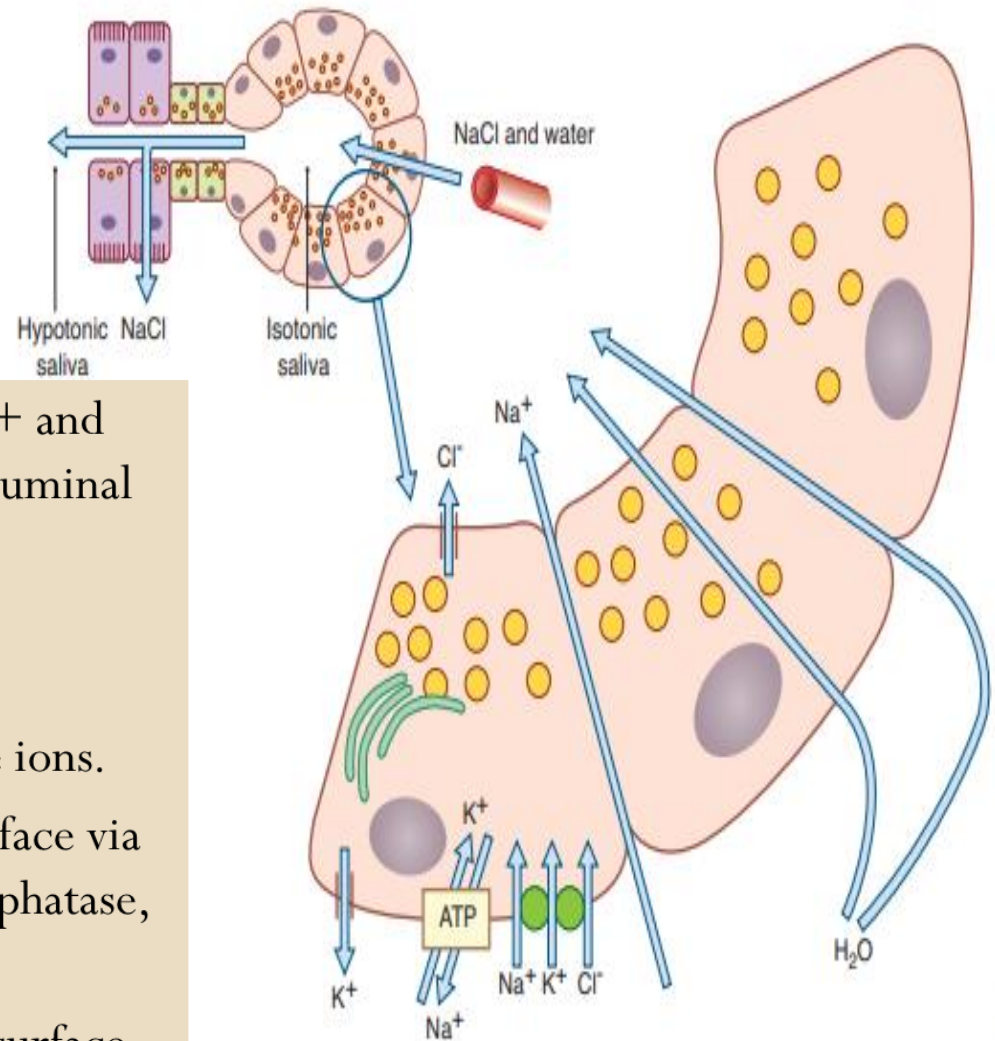


Fig
in:

AGE CHANGES

- EFFECTS ON SALIVARY FLOW:
 - One theory suggest little or no loss of salivary function.
 - Other stated that salivary secretion is in normal range during unstimulated stage, while less than normal during stimulation.

DISEASES

- Several viruses like EBV, HHV-6 & 7, cytomegalovirus & bacteria causes diseases of salivary gland.
- Mumps
- Bacterial infections



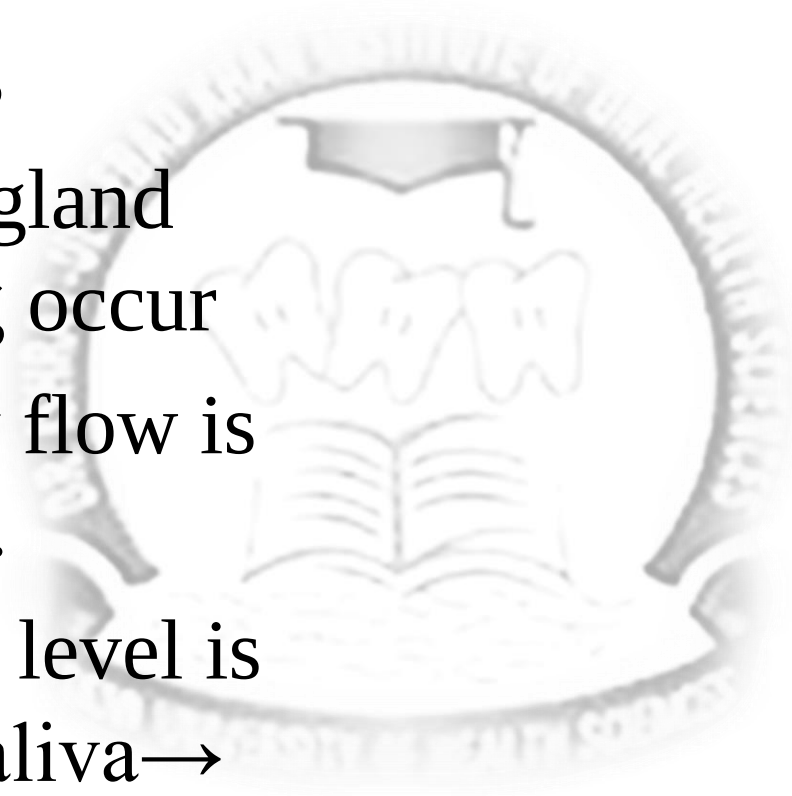
DISEASES

Blockage of duct can occur b/c of

- Sialoliths
- Mucous plugs
- Severing of duct of minor salivary gland by trauma .
- Also affected by benign & malignant neoplasm.

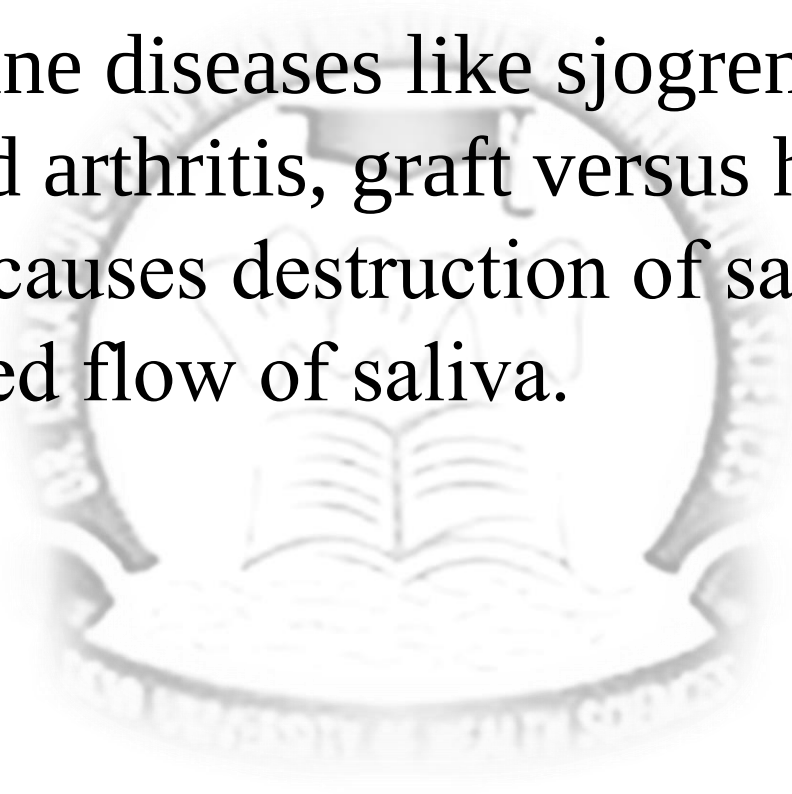
SYSTEMIC DISEASES

- In diabetes
 - Parotid gland swelling occur
 - Salivary flow is reduced.
 - Glucose level is ↑es in saliva → influences plaque metabolism.



SYSTEMIC DISEASES cont..

- Autoimmune diseases like sjogren syndrome, rheumatoid arthritis, graft versus host disease→ causes destruction of salivary tissue & ↓ed flow of saliva.

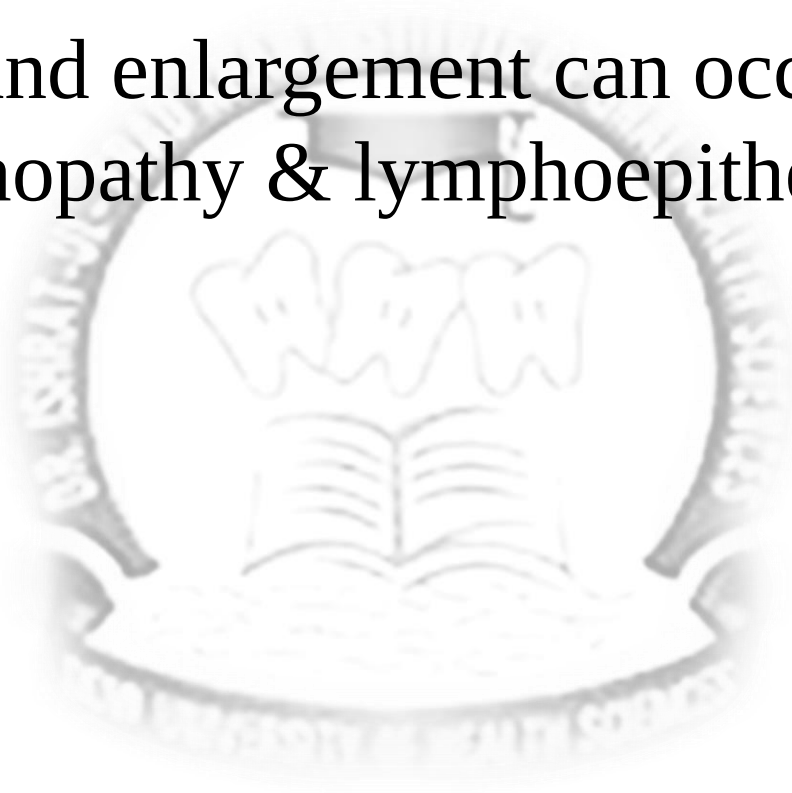


SYSTEMIC DISEASES CONT...

- In adrenal disease→ electrolyte composition is altered.
- AIDS→ salivary flow rate & level of secretory Ig.
- Cystic fibrosis→ salivary Na & Cl conc. Are ↑ed. Mucous secretory glands may develop mucous plugs.

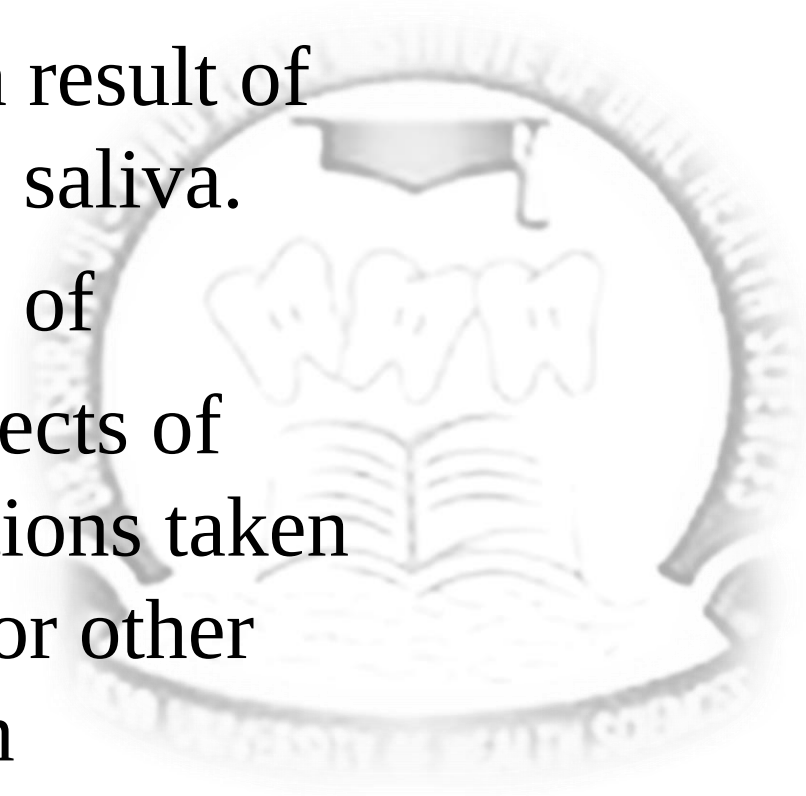
SYSTEMIC DISEASES cont..

- Parotid gland enlargement can occur b/c of lymphadenopathy & lymphoepithelial cyst.



DRY MOUTH (XEROSTOMIA)

- Occur as a result of little or no saliva.
- Occur b/c of
 - Side effects of medications taken by pts for other problem
 - Destruction of salivary gland (by radiotherapy of



DRY MOUTH (XEROSTOMIA)

- ↓ed volume leads to
 - drying of oral tissue & loss of protection
 - ↑ susceptibility to infection.
 - Difficulty in speech, eating & swallowing.
 - Teeth become susceptible to caries, esp. near the gingival margin.

DRY MOUTH (XEROSTOMIA)

- TREATMENT:
- Temporary relief is achieved by frequent sips of water or artificial saliva.
- Pts who have functioning salivary tissues may benefit from oral parasympathetic

